

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

1. Use Case Diagram

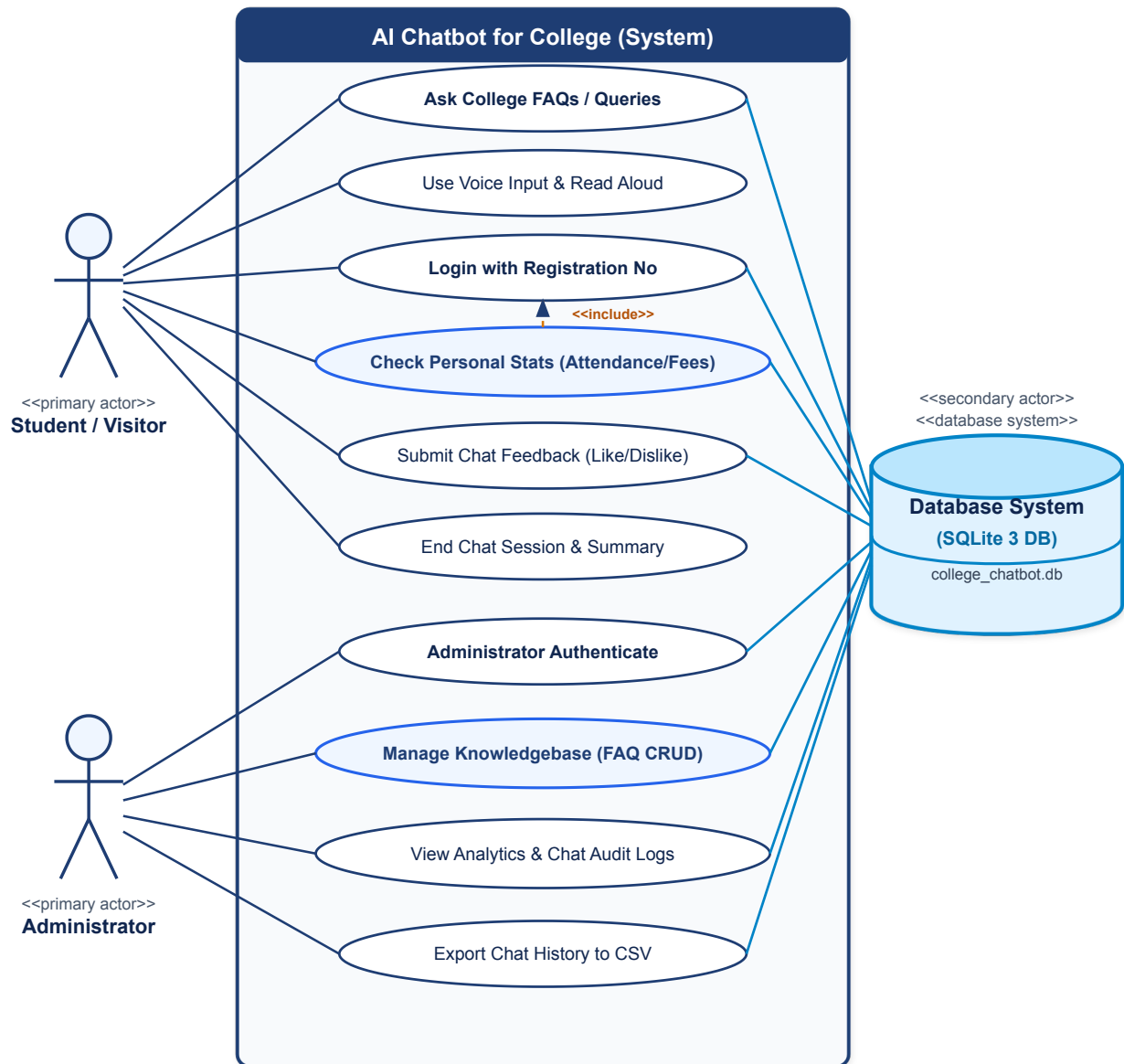


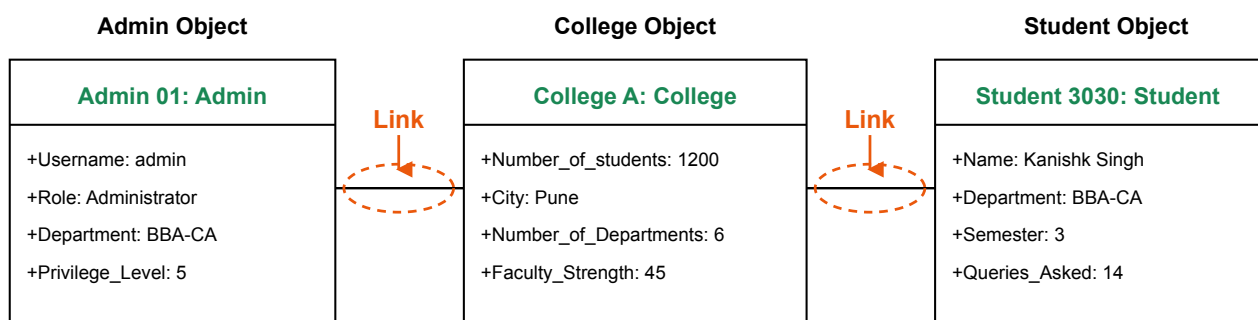
Fig 1: Use Case Diagram — primary actors (Student & Administrator), system boundary, and broad secondary database cylinder (SQLite 3 DB) with data-tier associations.

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

3. Object Diagram (Runtime Instances & Slot Values)

An object diagram using a link and 3 objects



An object of class Admin and an object of class Student are linked to an object of class College.

Object Diagrams | Unified Modeling Language (UML)

GfG

Fig 3: Object Diagram — runtime object instances (Admin Object, College Object, Student Object) connected via formal UML links matching GeeksforGeeks standard notation.

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

6. Data Flow Diagram (DFD Level 0 Context & Level 1 Detailed)

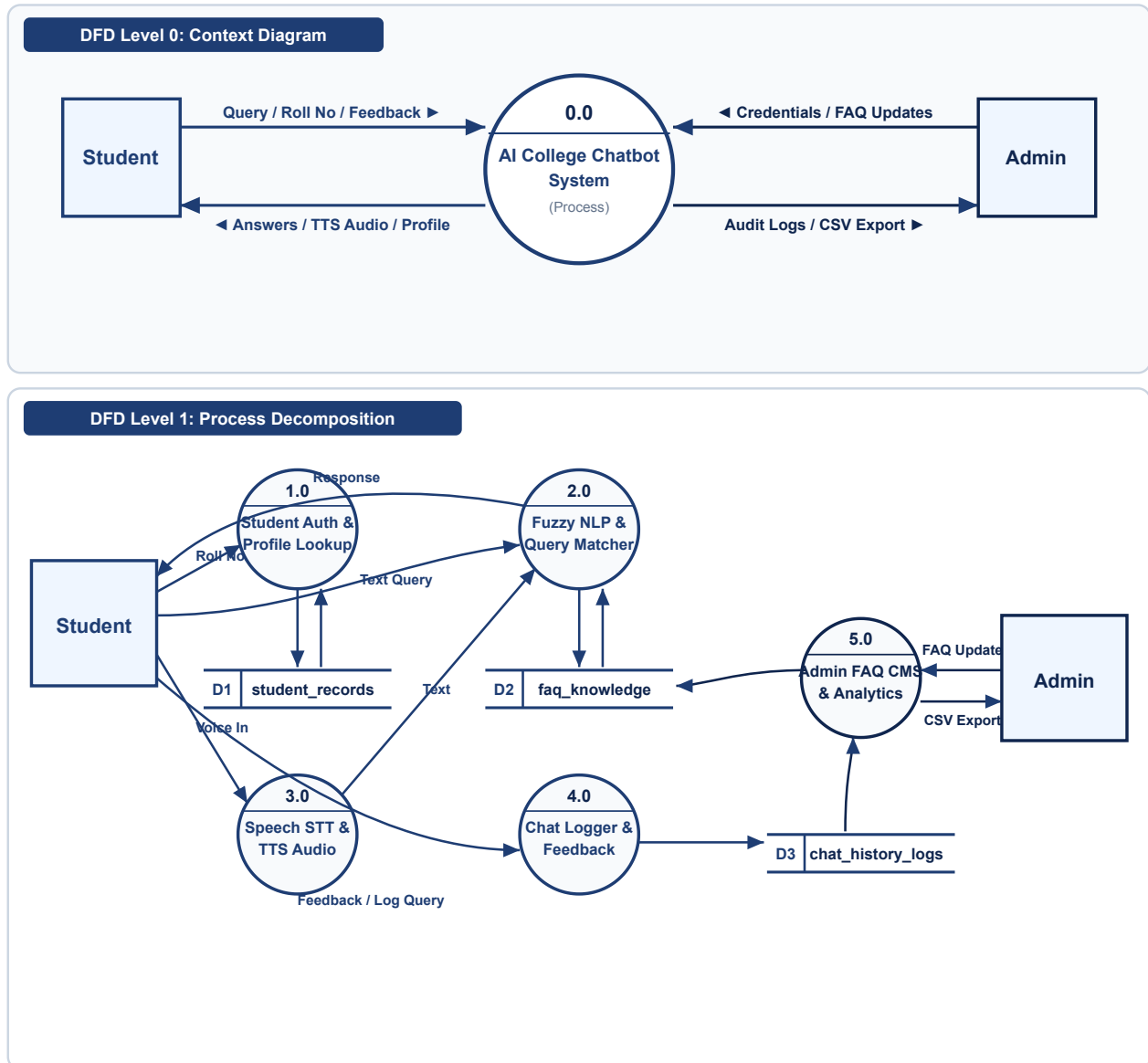


Fig 6: Data Flow Diagram (DFD) — Level 0 Context & Level 1 Process Breakdown using square entity symbols, circle process symbols, and solid data flow arrows.

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

7. Entity-Relationship Diagram (ERD — Database Architecture)

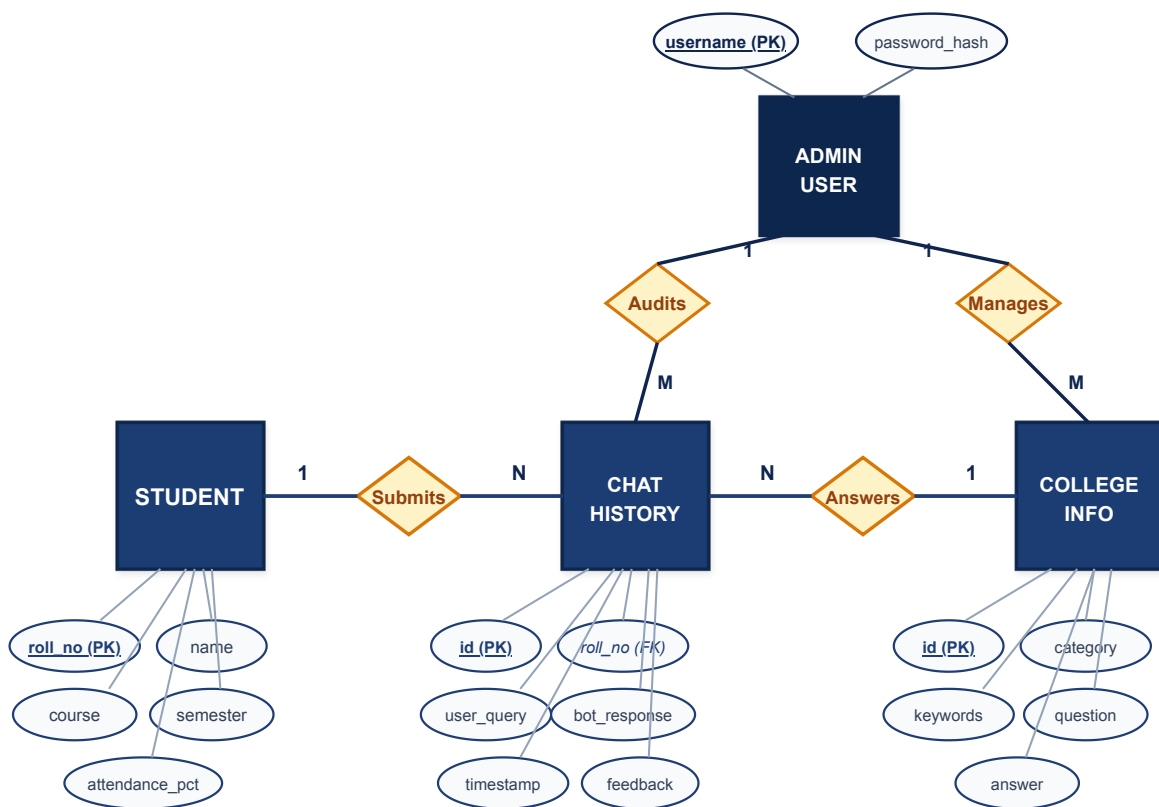


Fig 7: Entity-Relationship Diagram (ERD) — structured relational flow using square entity symbols (STUDENT, ADMIN_USER, CHAT_HISTORY, COLLEGE_INFO) with cardinalities and non-crossing attributes.

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

8. Deployment Diagram (Hosting Topology & Artifact Architecture)

deployment Development Environment Model

Development Environment Model

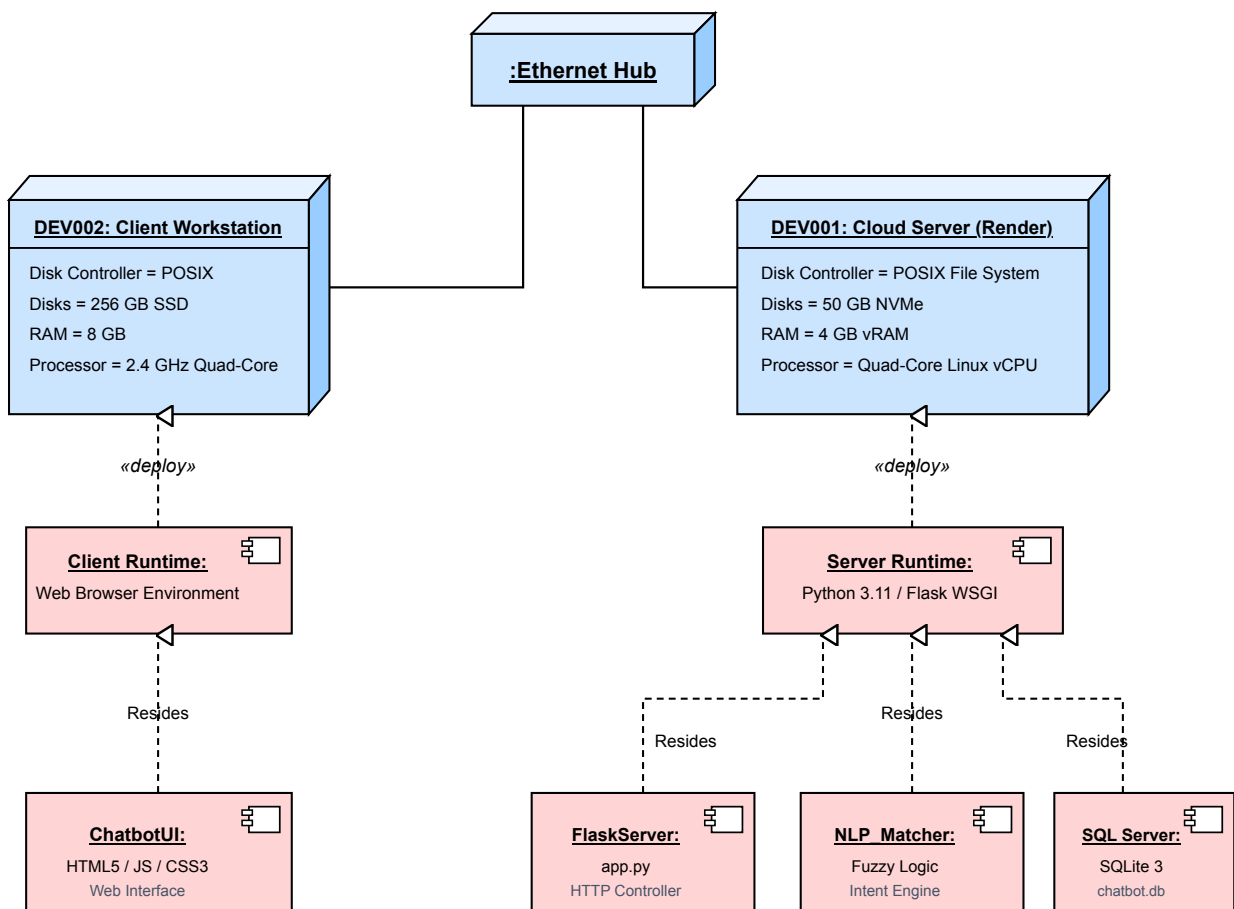


Fig 8: Deployment Diagram — Development Environment Model with :Ethernet Hub, 3D hardware nodes with specifications, «deploy» dependency arrows, and resident software component boxes.